

Hetul Mehta

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EDUCATION

University of California, San Diego

Master of Science in Computer Science | Artificial Intelligence | GPA: 4.00 / 4.00

San Diego, CA

09/2024 - 06/2026

- Teaching Assistant, CSE 193: Introduction to Computer Science Research | Mentor, Early Research Scholars Program

K. J. Somaiya College of Engineering

Bachelor of Technology in Information Technology | CGPA: 9.02 / 10.00

Mumbai, India

08/2019 - 05/2023

- Technical Head and Founder of **DataZen** (1st Data Science Club in Somaiya University), Research in **Machine Learning** (IEEE publication)
- Mentored 50+ undergraduate students in a year-long departmental research program, supporting completion of end-to-end research projects.

SKILLS

Programming & Web Technologies: Python, Java, JavaScript, TypeScript, SQL, GraphQL, HTML, CSS, React, Angular, Node.js, Express, Django, MongoDB, PostgreSQL, PHP, Flask

Cloud & Distributed Systems: AWS (Lambda, SNS, S3, DynamoDB, Bedrock Agents, OpenSearch), Redis, REST APIs, Apache Kafka, RabbitMQ, Microservices, Serverless Architecture

AI & Machine Learning: Gen AI, LLM, Agentic AI, Retrieval-Augmented Generation, TensorFlow, Keras, scikit-learn, Computer Vision

Certifications: AWS Cloud Practitioner

WORK EXPERIENCE

Amazon

Seattle, WA

Software Development Engineer Intern

06/2025 - 09/2025

- Architected an **AI-powered validation system** using **Amazon Bedrock Agents** and **AWS Lambda**, processing **77K+ promotional messages weekly**, resolving production issues where unclear messages had caused **customer confusion and support escalations**
- Developed **RAG knowledge base** with **OpenSearch vector store and S3**, leveraging **20K+ production samples** to enable **context-aware validation** where no review process previously existed, improving message clarity and preventing customer-facing errors
- Orchestrated an **event-driven, serverless pipeline** integrating Bedrock Agents, Lambda, S3, and SNS, **blocking 68 problematic messages in the first week of production**, automatically raising tickets for promotion owners and preventing unclear promotions from reaching customers

JP Morgan Chase

Mumbai, India

Software Engineer

08/2023 - 08/2024

- Delivered a **React-based UI form builder**, enabling rapid creation and deployment of responsive forms across micro-frontends, reducing development time by **70%**.
- Implemented a **Kafka-based high-throughput messaging system**, automating case creation and processing **10K+ daily messages** with **99.9% accuracy**, significantly improving operational efficiency.
- Designed **configurable entitlements middleware** using React and Java with data manipulation and masking capabilities, resulting in **50% faster policy updates** and **90% reduction in unauthorized data exposure**.

Software Engineer Intern

06/2022 - 08/2022

- Automated database table comparison for regression testing, developing an algorithm that replaced manual Excel-based validation and **reduced comparison time by 80%**.

Department of Information Technology, KJSCE

Mumbai, India

Research & Development Intern, Machine Learning Research

10/2020 - 03/2022

- Led the **"Domain Classification for the Web"** project, building an ML model to categorize websites into **16 classes** with **94% accuracy**, dataset released on Kaggle (5500+ downloads).
[Dataset](#), [Notebook](#)
- Proposed a **keyword-based retrieval algorithm** for Dark Web analysis, leading to publication of **"Keyword-Based Information Retrieval Model for the Dark Web"** in an **IEEE journal**.
[IEEE](#)
- Performed **community detection analysis** on Dark Web networks using **Leiden**, **Louvain**, and **Label Propagation**, identifying **100+ communities** with high modularity.
[GitHub](#)

Levyne

Mumbai, India

Backend Development Intern, Startup

02/2021 - 08/2021

- Integrated **live order tracking** and **courier API** into the Levyne application and contributed to the development of the Levyne Dashboard, improving logistics visibility and customer experience.
- Developed **web scraping pipelines** to aggregate competitive pricing data from major e-commerce platforms, processing information for **100K+ products** to support pricing and market analysis.

PROJECTS

SC-SMOTE (Semantic-Cosine based Synthetic Minority Oversampling Technique)

[GitHub](#), [PyPI](#)

- Conceived and prototyped a **novel algorithm** for **synthetic textual data generation**, boosting classification performance on imbalanced datasets by up to **20%**.
- Published and released as an open-source Python package for seamless adoption in machine learning pipelines.

StyleWise

[GitHub](#)

- Built a **fashion recommendation system** using **TensorFlow** and **Computer Vision**, achieving **80% accuracy** on product matching tasks.
- Engineered an **image processing pipeline** to extract features from **10K+ clothing images**, powering similarity-based recommendations.